Electronic Supplementary Information (ESI)

Excitation energy dependence of Raman spectra of few-layer WS_2

Jinho Yang, Jae-Ung Lee and Hyeonsik Cheong

Department of Physics, Sogang University, Seoul 04107, Korea

E-mail: hcheong@sogang.ac.kr.

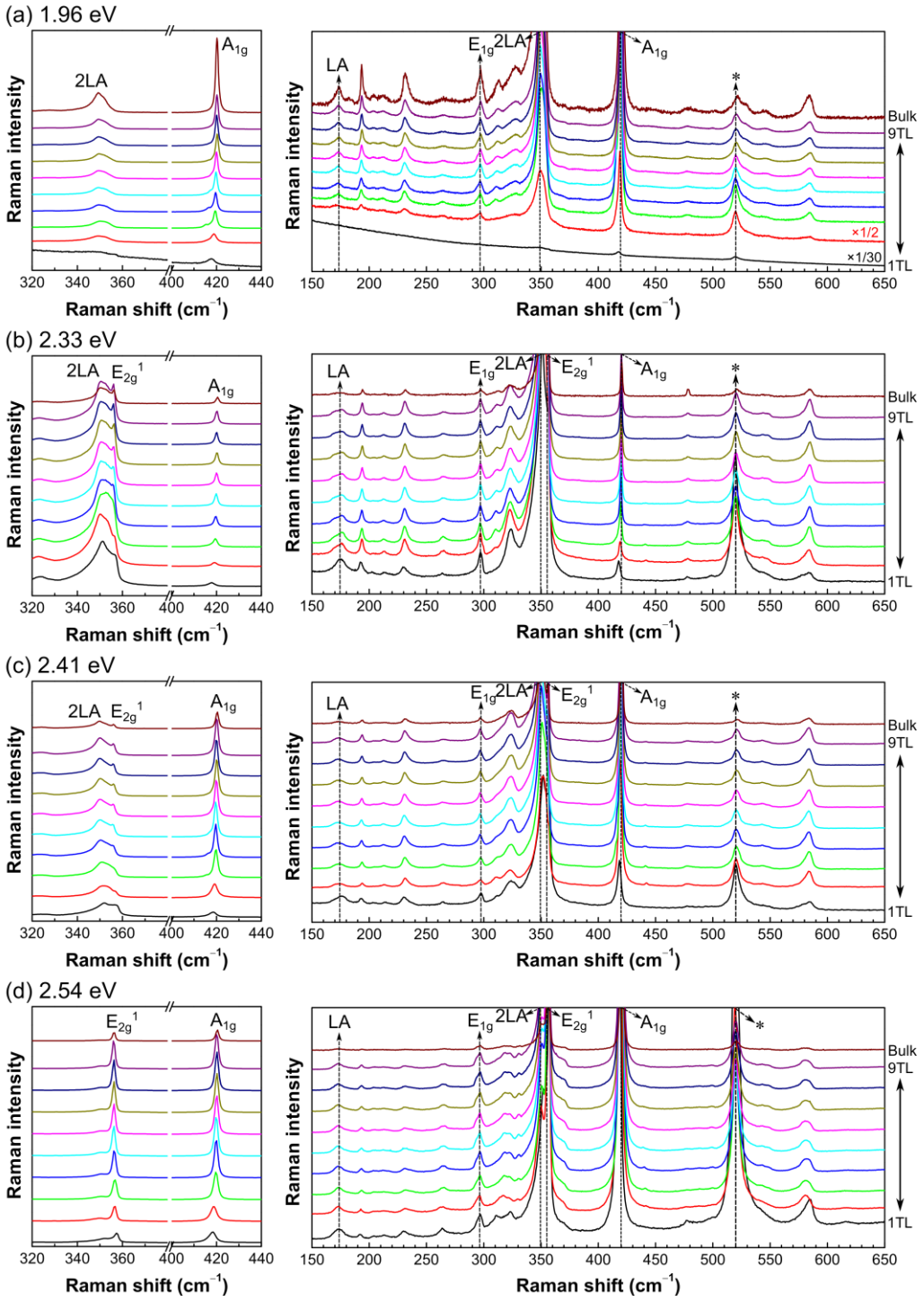


Fig. S1. Thickness dependence of Raman spectra for excitation energies of (a) 1.96, (b) 2.33, (c) 2.41 and (d) 2.54 eV. The mode notations are based on the case of bulk WS_2. Silicon peak (520 cm^{-1}) is indicated by '*'.

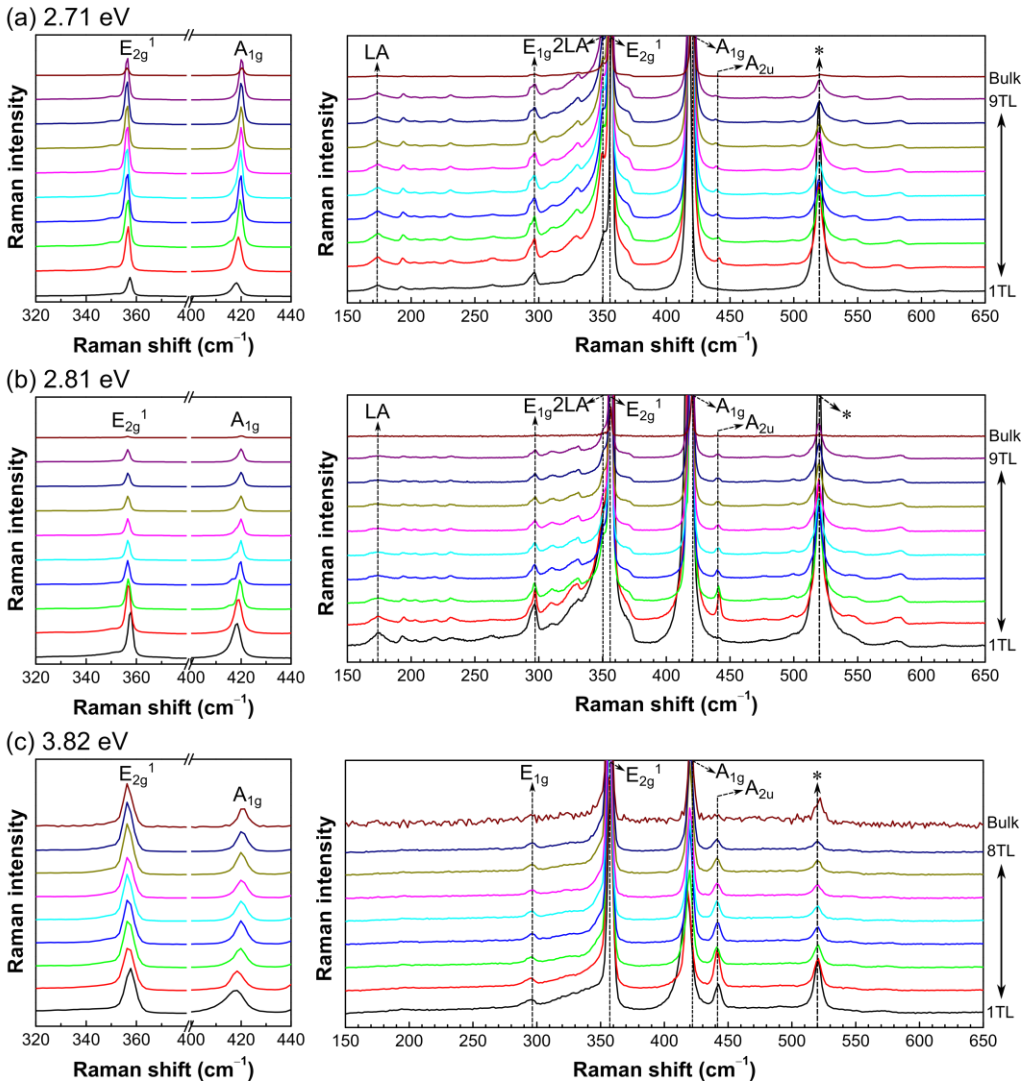


Fig. S2. Thickness dependence of Raman spectra for excitation energies of (a) 2.71, (b) 2.81, and (c) 3.82 eV. The mode notations are based on the case of bulk WS_2. Silicon peak (520 cm^{-1}) is indicated by '*'.

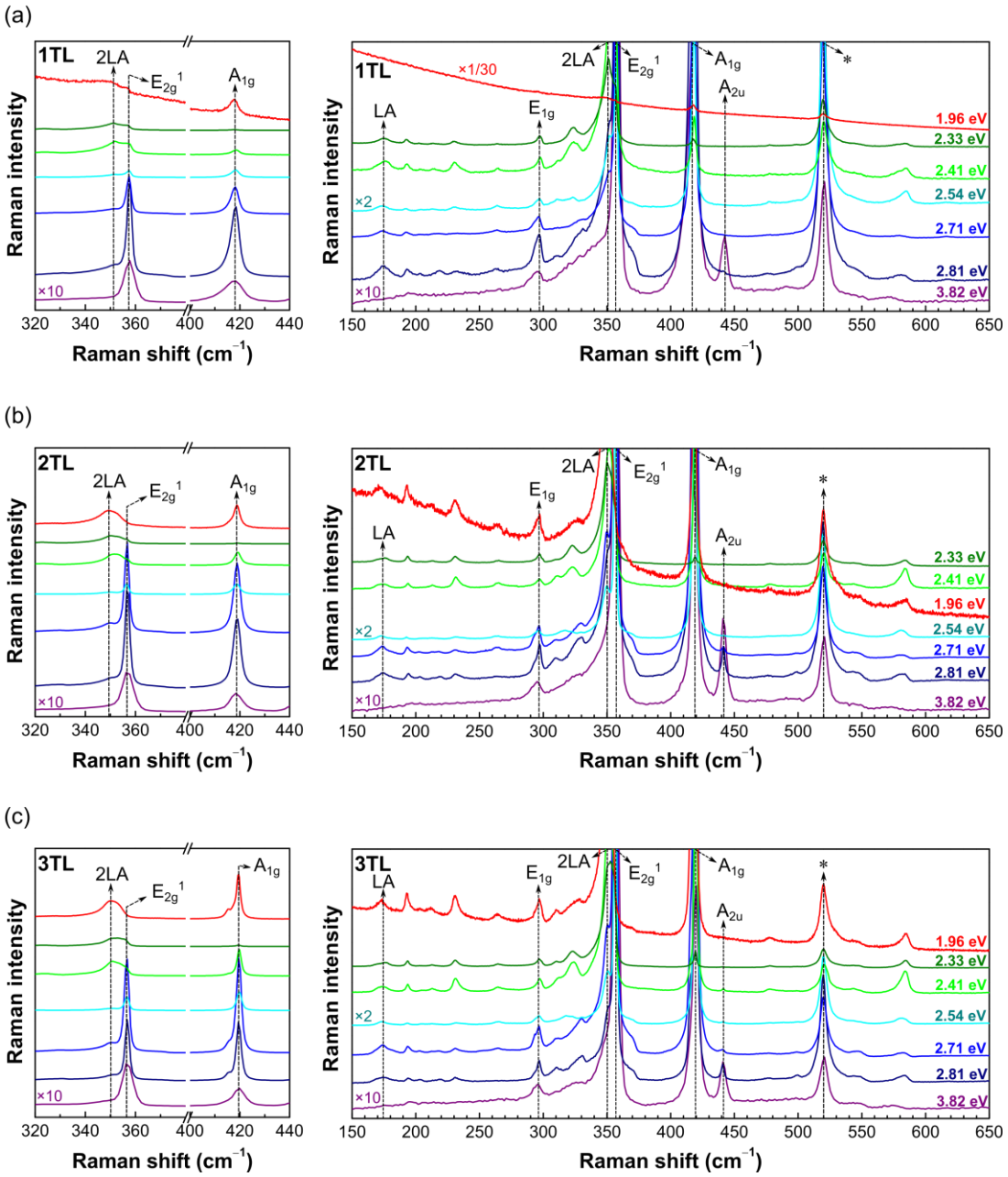


Fig. S3. Excitation energy dependence of Raman spectra of (a) 1TL, (b) 2TL, and (c) 3TL WS_2.

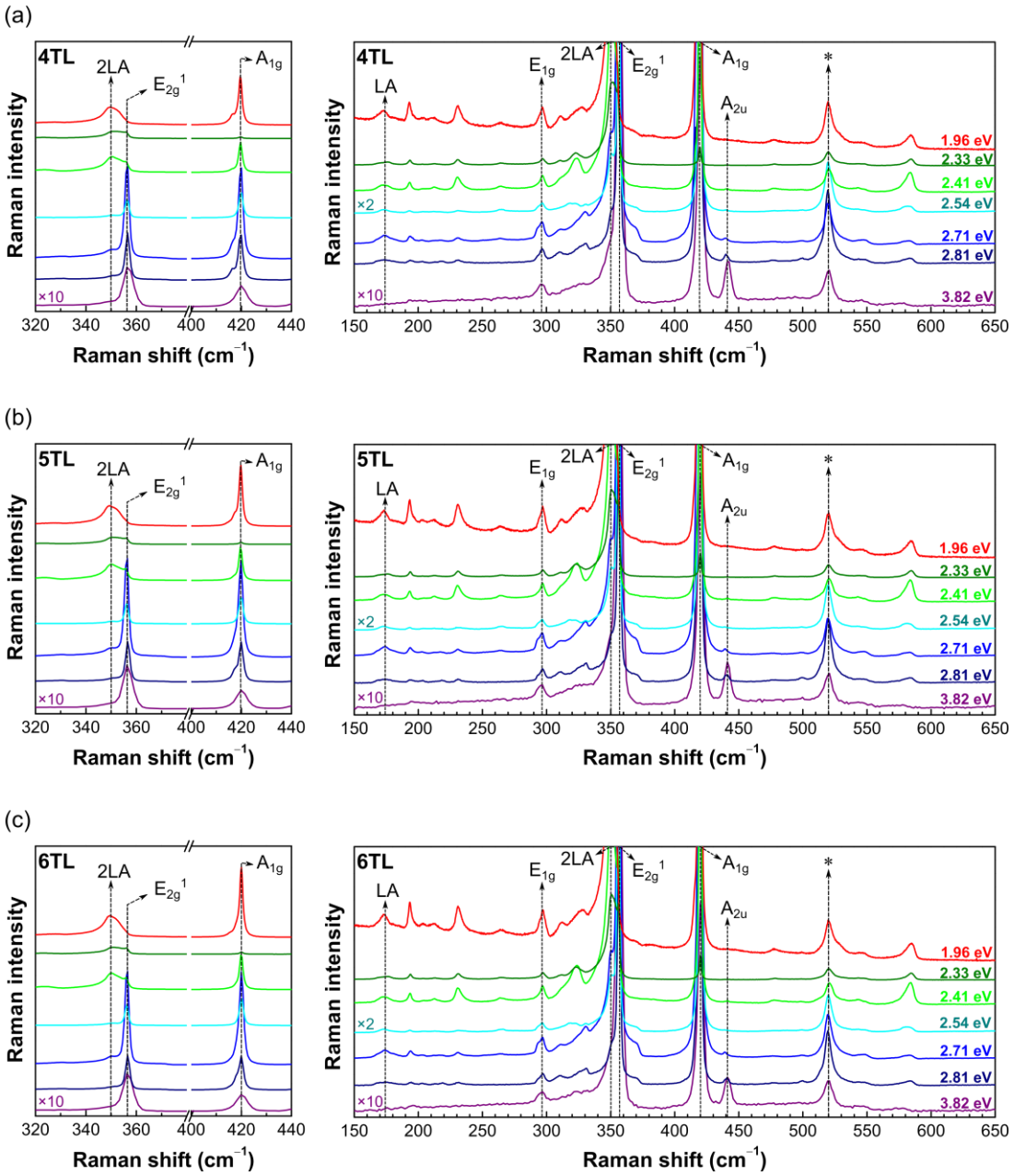


Fig. S4. Excitation energy dependence of Raman spectra of (a) 4TL, (b) 5TL, and (c) 6TL WS_2.

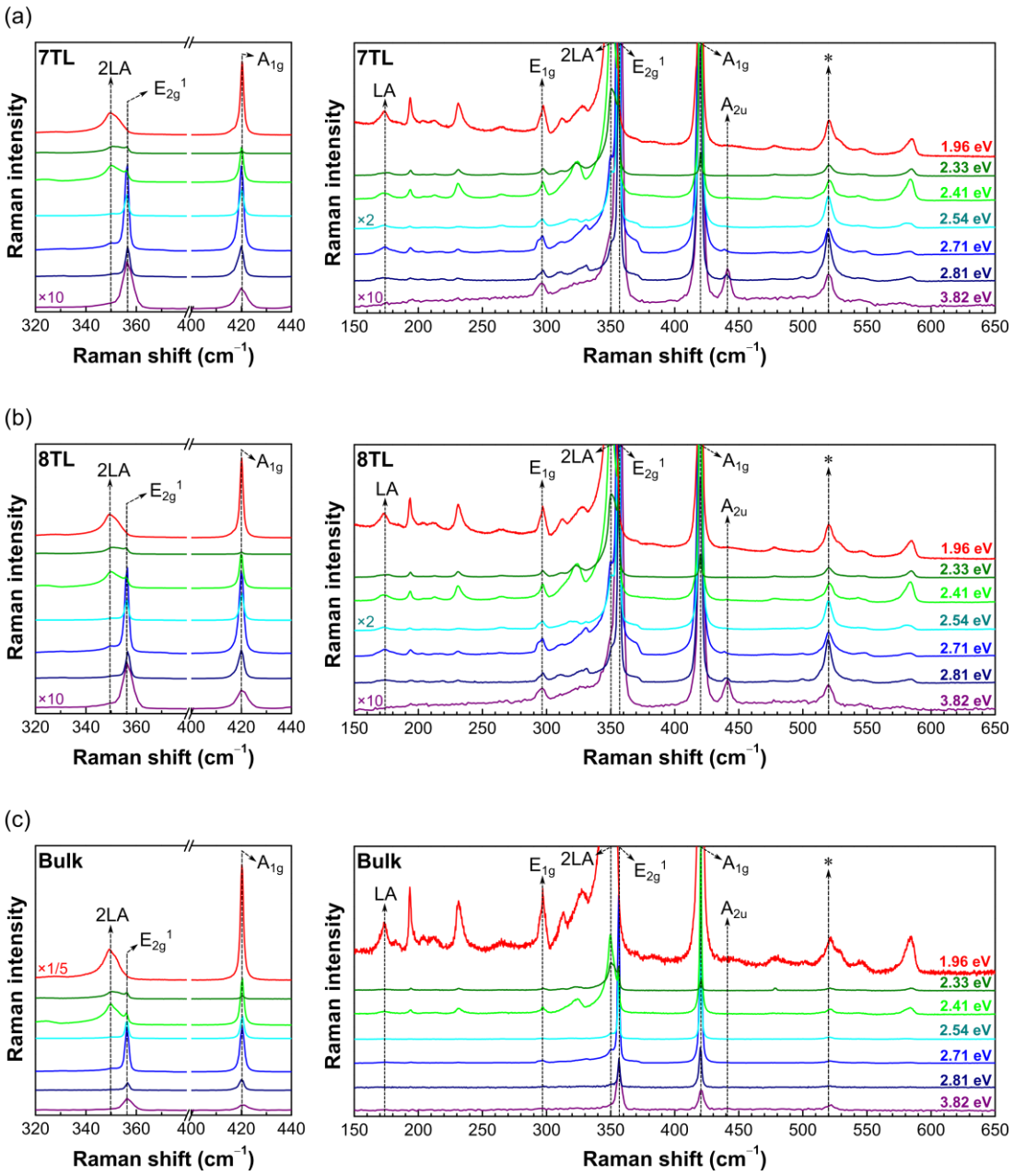


Fig. S5. Excitation energy dependence of Raman spectra of (a) 7TL, (b) 8TL, and (c) bulk WS_2.

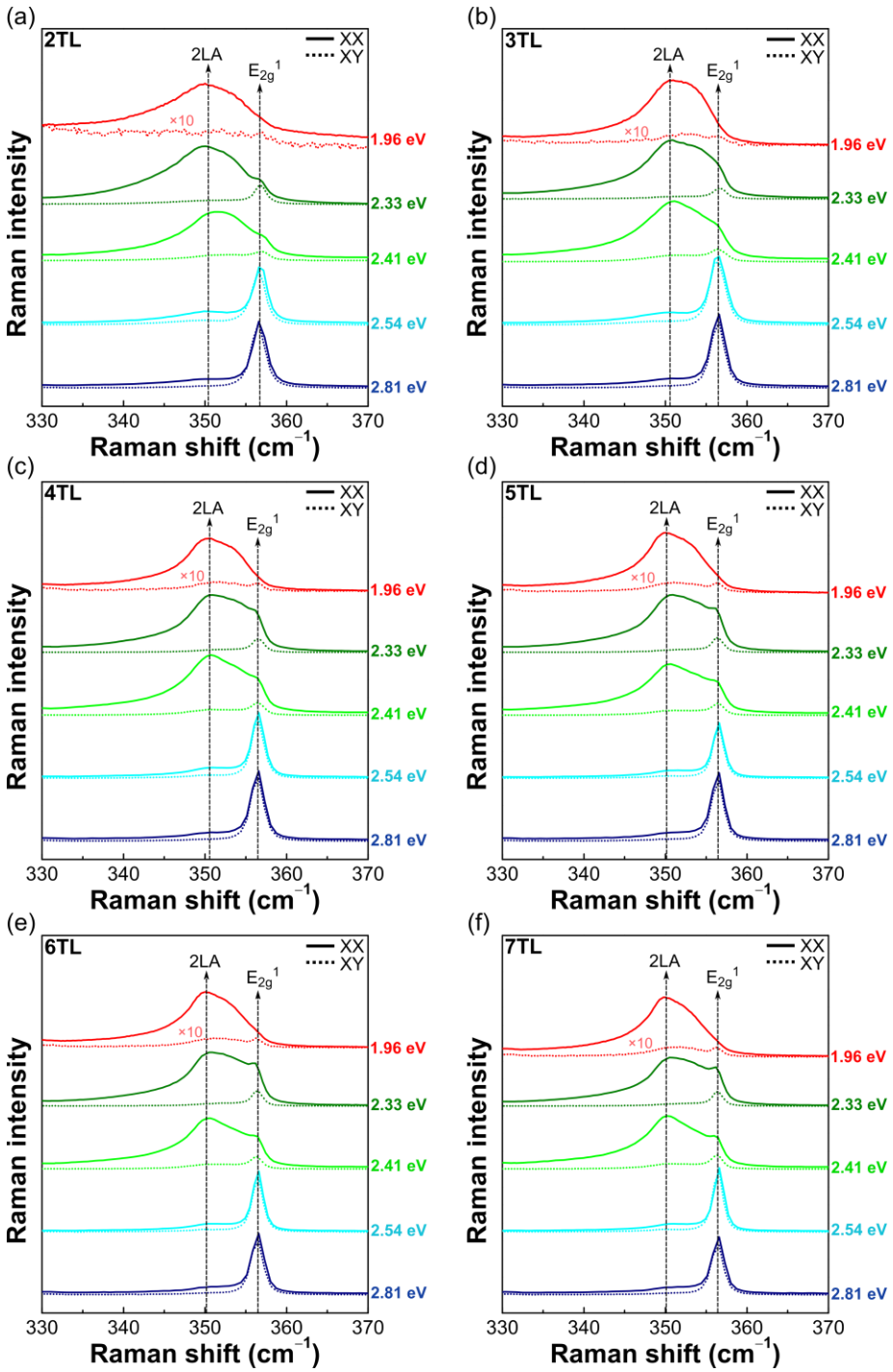


Fig. S6. Polarized Raman spectra of the region near the E_{2g}^1 mode for 2 to 7 TL WS_2 for five different excitation energies. Solid curves are for parallel polarization (XX), and dashed lines are for cross polarization (XY).

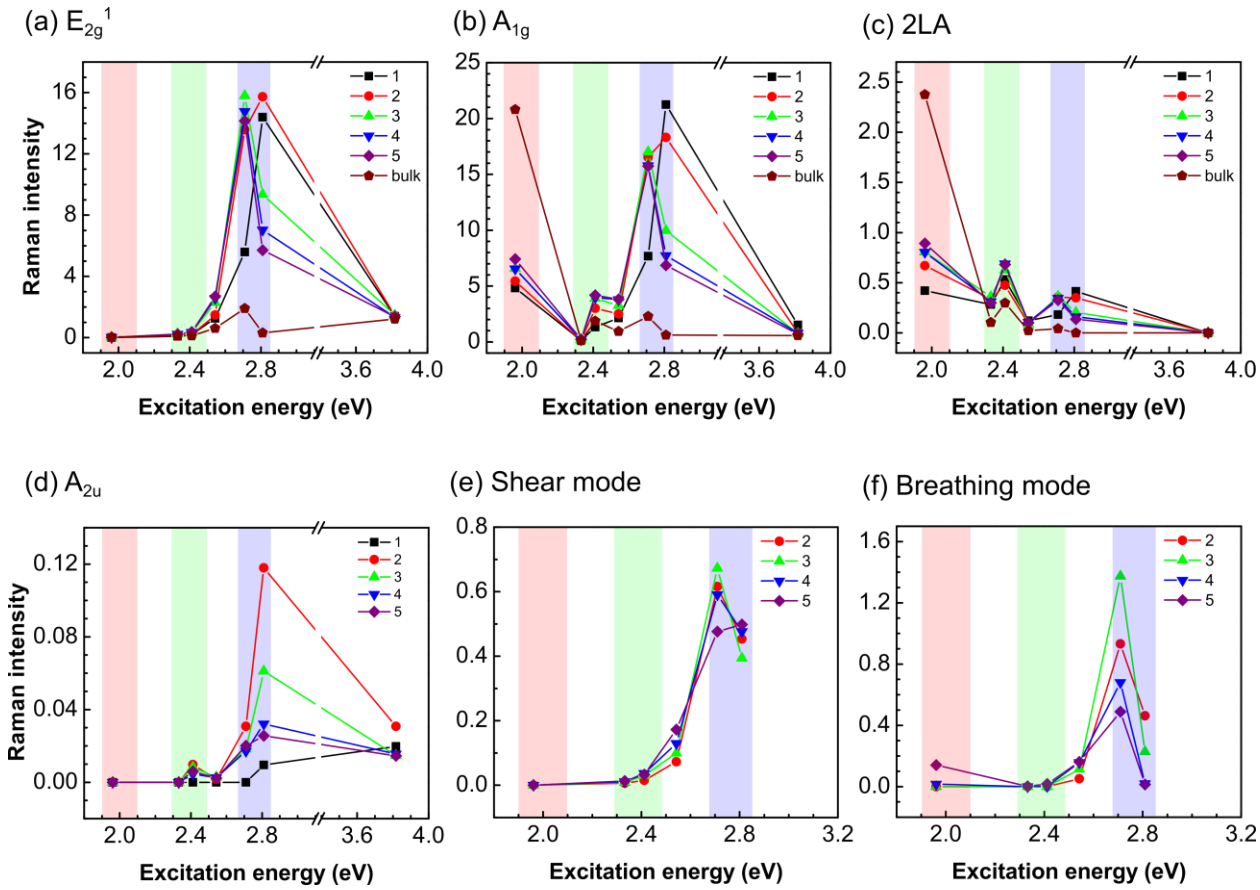


Fig. S7. Resonance profiles of six Raman modes for 1 to 5TL and bulk WS_2. The approximate energies of A, B, and C exciton states at ~2.0, ~2.4, and ~2.8 eV, respectively, are indicated.

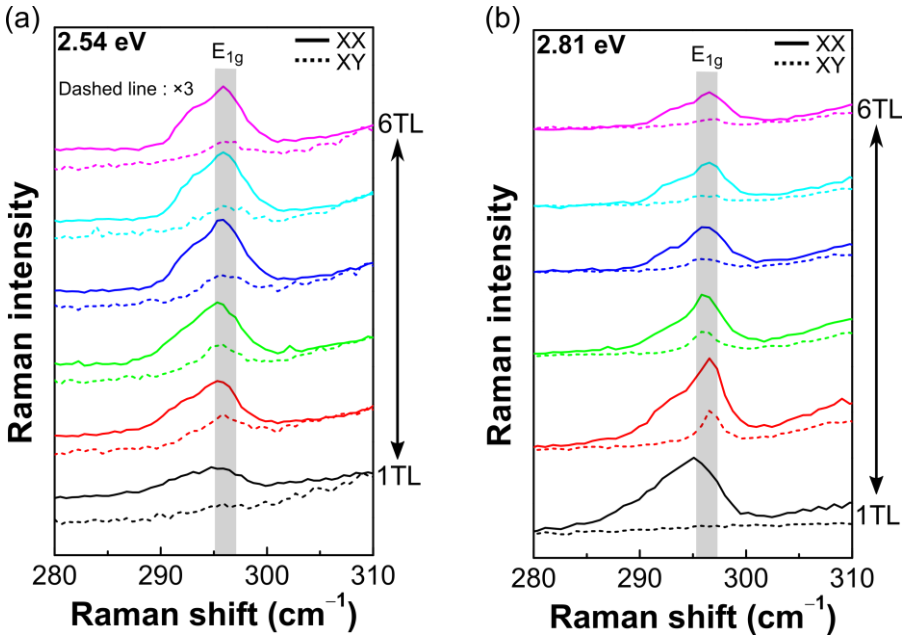


Fig. S8. Polarized Raman spectra of E_{1g} peak measured with (a) 2.54 and (b) 2.81 eV excitation energies. The E_{1g} peak is identified in cross polarization (XY) for 2TL to 6TL but does not appear for 1TL.

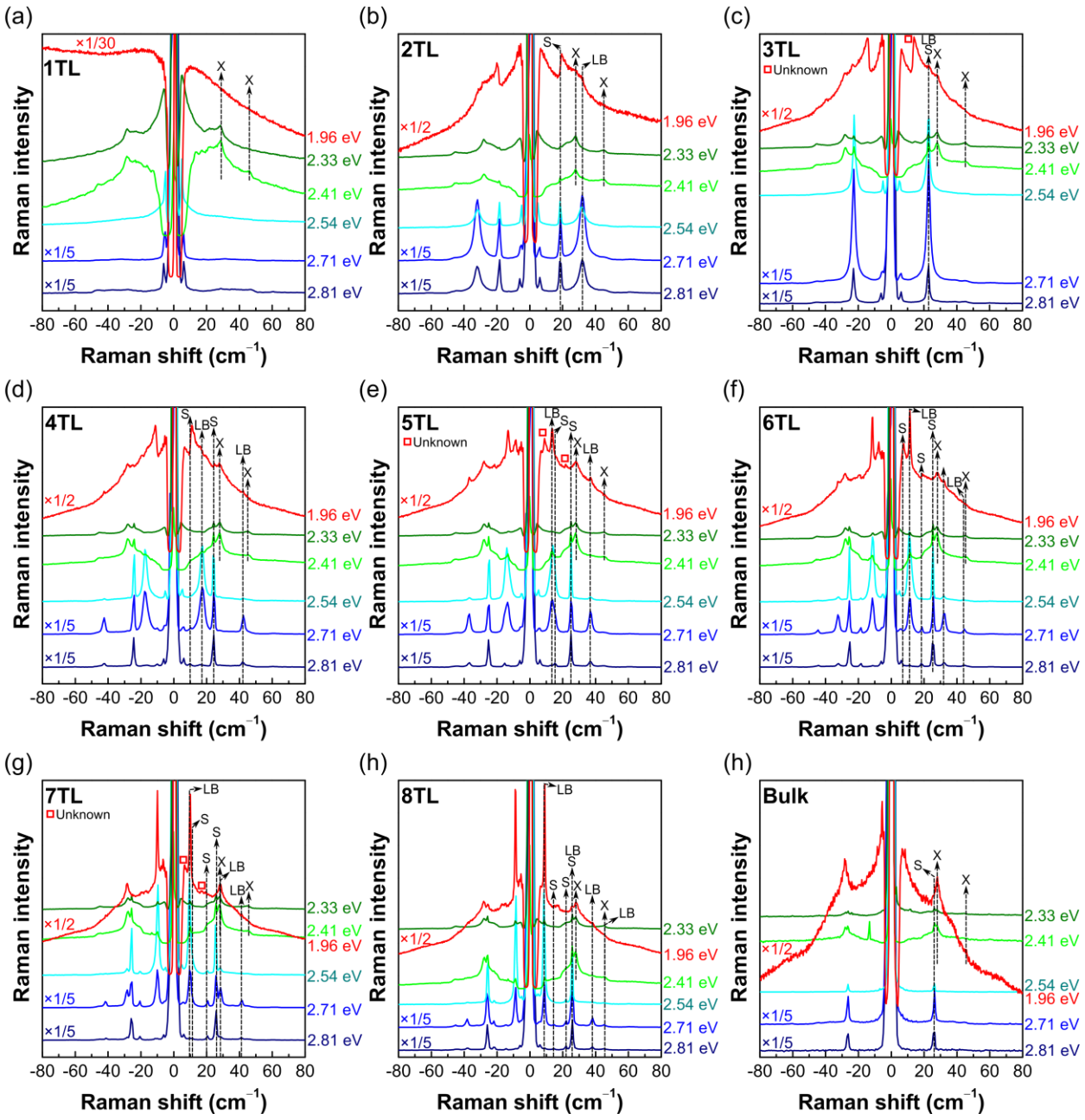


Fig. S9. Excitation energy dependence of Raman spectra of 1TL to 8TL and bulk WS_2 in the low frequency region.

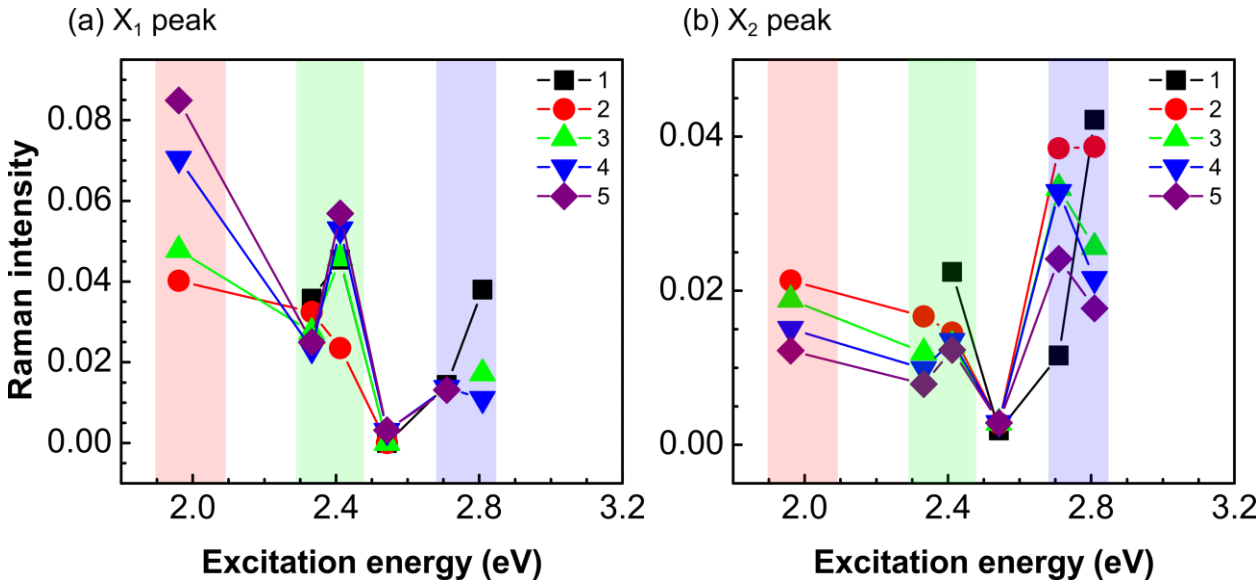


Fig. S10. Resonance profiles of two X peaks for 1 to 5TL and bulk WS_2. The approximate energies of A, B, and C exciton states at ~2.0, ~2.4, and ~2.8 eV, respectively, are indicated.

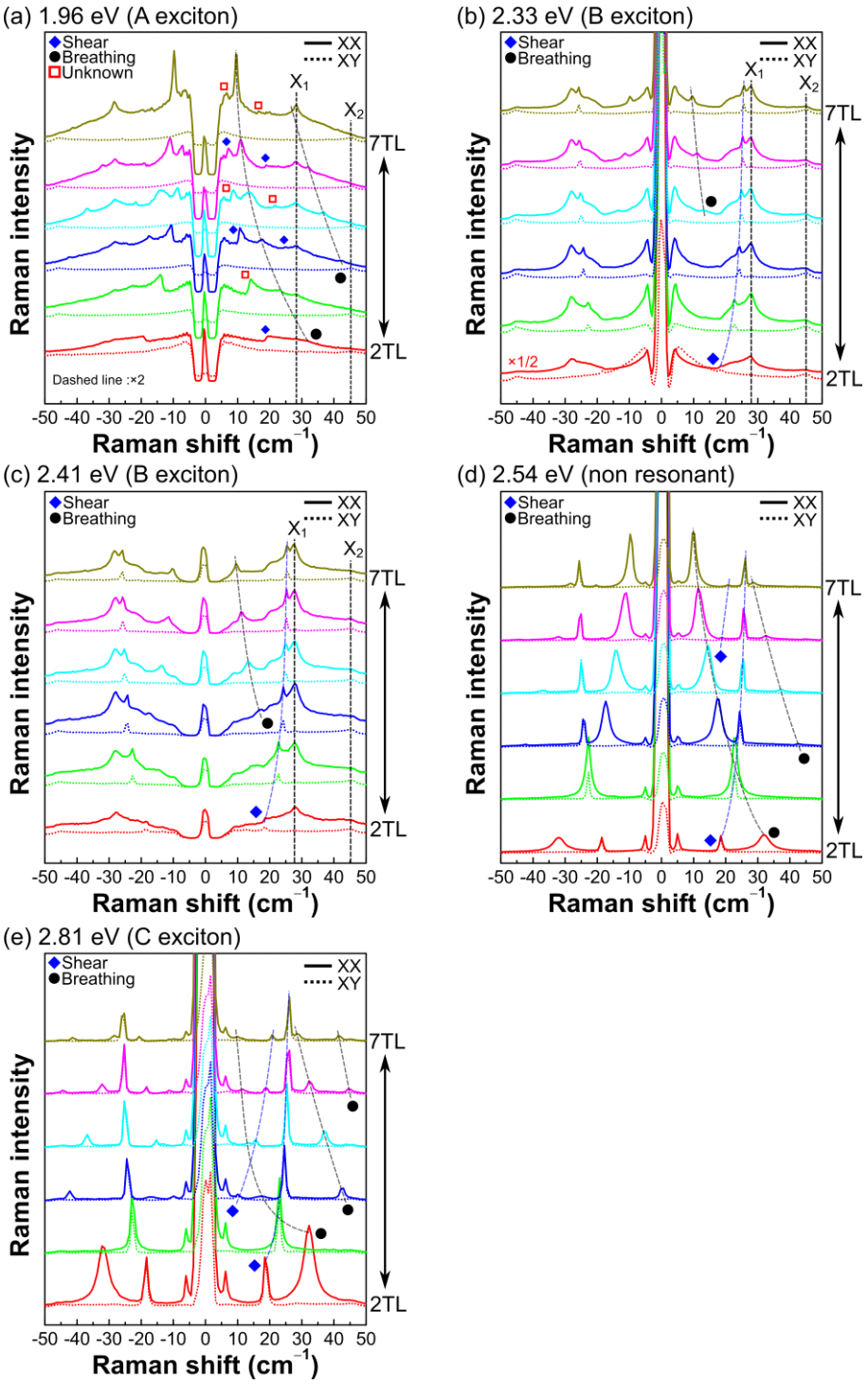


Fig. S11. Polarized Raman spectra in the low frequency region for 2TL to 7 TL WS_2 measured with five different lasers. Solid curves are for parallel polarization (XX), and dashed curves are for cross polarization (XY).

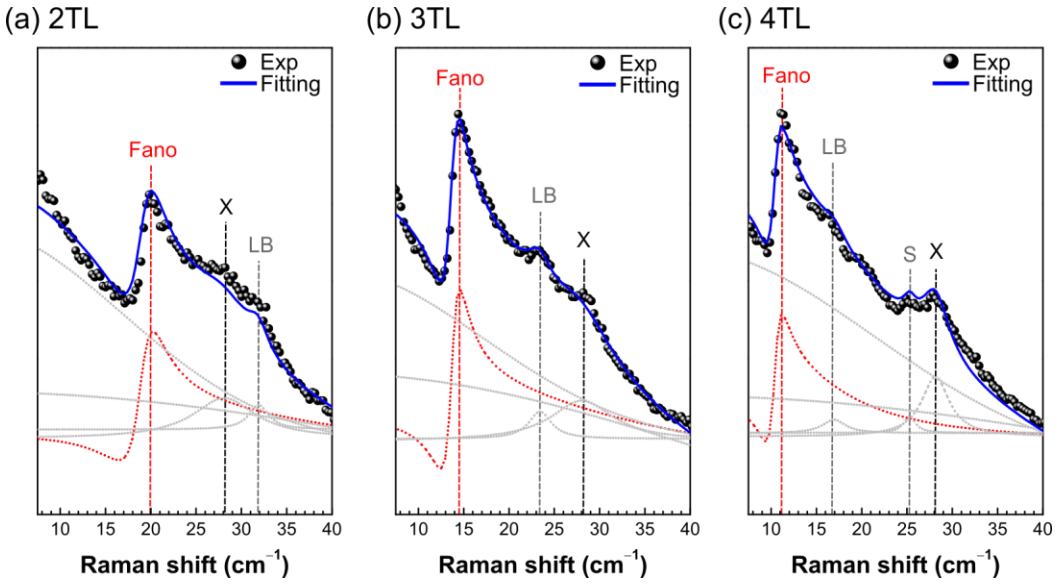


Fig. S12. Deconvolution of the low-frequency Raman spectra measured with 1.96 eV excitation. The lowest shear mode is fitted with the BWF line shape and the rest with Lorentzian line shapes.